If the side of a cube is 'a' units then the Volume of the cube (V) = a^3 cubic units.

Note

For any two cubes, the following results are true.

- Ratio of surface areas = (Ratio of sides)^2
- Ratio of volumes = (Ratio of sides)^3
- (Ratio of surface areas)^3 = (Ratio of volumes)^2

Example 7.13

Find the volume of cube whose side is 10 cm.

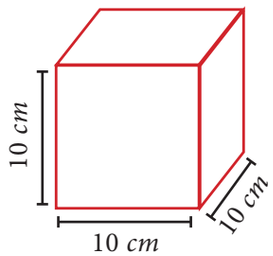


Fig. 7.26

Solution

Given that side (a) = 10 cm

volume of the cube $= a^3$

$= 10 \times 10 \times 10$

$= 1000\ cm^3$

Example 7.14

A cubical tank can hold 64,000 litres of water. Find the length of its side in metres.

Solution

Let 'a' be the side of cubical tank.

Here, volume of the tank $= 64{,}000$ litres

$$\text{i.e., } a^3 = 64{,}000 = \frac{64000}{1000} \quad [\text{since, } 1000 \text{ litres} = 1m^3]$$

$$a^3 = 64\ m^3$$

$$a = \sqrt[3]{64} \quad a = 4\ m$$

Therefore, length of the side of the tank is 4 metres.

Example 7.15

The side of a metallic cube is 12 cm. It is melted and formed into a cuboid whose length and breadth are 18 cm and 16 cm respectively. Find the height of the cuboid.

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